

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

QIAN QI

ABSTRACT. The raw perspective-entropy Hessian is not assumed positive. We prove short-time coercivity directly on the balanced tangent by a weighted linearized Boltzmann energy estimate, construct covariance first from exact finite-volume cumulants, and identify its inverse with the second epi-derivative of the exposed rate. A backward observability theorem closes the full balance gauge and localized connected cumulants give the prepared density–contact Gaussian process.

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1. ROUND-TEN RECONSTRUCTION: COVARIANCE FIRST, SHORT-TIME CONSTRAINED CURVATURE, AND THE COMPLETE GAUGE

The collision action is not assigned a Riemannian metric by formal inspection. Its raw second variation may be indefinite. We first construct the Gaussian covariance from exact finite-volume cumulants. We then prove a short-time energy estimate for the actual linearized balanced dynamics and identify the second epi-derivative of the exposed rate with the inverse covariance on its range. No fictitious curvature is attributed to the linear balance constraint.

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1.1. The weighted balance complex. Fix $m > 8$. Let $\mathbf{X}_T$ be the closure of smooth density tangents for

$$\|u\|_{\mathbf{X}_T}^2 = \sup_{t \leq T} \int \frac{|u_t|^2}{f_t} (1 + |v|^m) + \int_0^T \|u_t\|_{H_{x,v}^{-1}(w_m)}^2 dt,$$

and let $\mathbf{Y}_T = L^2(qA_f)$ be the contact tangent space. Write a contact tangent as

$$J = q DA_f[u] + qA_f \xi.$$

The linearized weak balance is

$$\mathcal{B}_{f,q}(u, \xi) = \partial_t u + v \cdot \nabla_x u - \Delta^*(qDA_f[u]) - \Delta^*(qA_f \xi) = 0,$$

with the prepared initial conservation constraints. Its closed kernel is $\mathbf{T}_{f,q}$.

The raw second variation of the exposed action is

$$\mathfrak{h}_{f,q}(u, \xi) = D^2 I_{0,a}(f_0)[u_0, u_0] + \int_0^T \int qA_f |\xi|^2 + \int_0^T \int (1-q) D^2 A_f[u, u].$$

The last integral has no fixed sign.

Theorem 1.1 (Linearized biased Boltzmann well-posedness). *On every compact regular phase chart there is $T_* > 0$ such that, for $T \leq T_*$, the equation $\mathcal{B}_{f,q}(u, \xi) = 0$ with prepared initial datum has a unique solution in $\mathbf{X}_T$ and*

$$\|u\|_{\mathbf{X}_T}^2 \leq C \left(\|u_0\|_{L^2(f_0^{-1}w_m)}^2 + \int_0^T \int qA_f |\xi|^2 \right).$$

The same estimate holds for the backward adjoint, including its endpoint and B2 Green-trace terms.

Proof. Decompose u/f into the five collision invariants and their orthogonal complement in the weighted collision Hilbert space. On the complement, the symmetric part of the regular biased collision operator obeys the uniform spectral-gap estimate

$$-\langle h, L_{f,q}^{\text{sym}} h \rangle \geq c \|h\|_\nu^2.$$

The invariant coefficients solve the finite transport system obtained by integrating against $1, v, |v|^2$; their initial values are fixed by the microcanonical tangent constraints. Pair the equation with u/f after a velocity truncation. Free transport is skew after the endpoint terms, B2's closed Green graph cancels incoming and outgoing specular traces, and the forcing $\Delta^*(qA_f \xi)$ is bounded by Cauchy–Schwarz in the collision Dirichlet norm. The high-velocity remainder tends to zero by the m th moment. Gronwall on a compact coefficient chart gives the estimate and uniqueness. Reversing time and using the adjoint reflection law gives the backward estimate. $\square$

Corollary 1.2 (Positive short-time second epi-curvature). *After reducing T_* if necessary, the restriction of $\mathfrak{h}_{f,q}$ to $\mathbf{T}_{f,q}$ is coercive:*

$$\mathfrak{h}_{f,q}(u, \xi) \geq c \left(\|u_0\|_{L^2(f_0^{-1}w_m)}^2 + \|u\|_{\mathbf{X}_T}^2 + \int qA_f |\xi|^2 \right).$$

No curvature is added from the linear balance multiplier.

Proof. Bilinearity of A_f and the moment bounds give

$$\left| \int_0^T \int (1-q) D^2 A_f[u, u] \right| \leq CT \|q - 1\|_{\infty, w} \|u\|_{\mathcal{X}_T}^2.$$

Use Theorem 1.1 and choose T_* so that this term is absorbed by the positive initial entropy and collision residual. This is a direct constrained energy argument, not a KKT-curvature assertion. $\square$

1.2. Closed range and the balance gauge. Let $\mathbf{P}_T$ be the weighted backward transport Sobolev space with endpoint traces, and let

$$\mathfrak{D}(p, \psi) = \Delta p + \psi, \quad \mathfrak{G}r = (r, -\Delta r).$$

Constants and collision invariants are fixed by endpoint normalization.

Theorem 1.3 (Backward observability and exact gauge). *For every normalized cotangent,*

$$\|(p, \psi)\|_{\mathbf{P}_T \times L^2(qA_f)} \leq C \left(\|\mathcal{B}_{f,q}^*(p, \psi)\|_{\mathbf{T}_{f,q}^*} + \|\Delta p + \psi\|_{L^2(qA_f)} \right).$$

Consequently the balance operator has closed range and

$$\ker \mathfrak{D} = \text{Ran } \mathfrak{G}.$$

The cotangent space is the Hausdorff quotient $(\mathbf{P}_T \times L^2(qA_f))/\text{Ran } \mathfrak{G}$.

Proof. Solve the backward equation by the micro-macro decomposition used in Theorem 1.1. The collision gap controls the microscopic part, while the five macroscopic coefficients solve a finite hyperbolic system controlled by the endpoint normalization and the adjoint balance functional. The B2 trace graph supplies the boundary Green identity and excludes grazing defect functionals. The resulting observability estimate implies closed range by the Hilbert closed-range theorem. If $\Delta p + \psi = 0$, the pair equals $(p, -\Delta p) = \mathfrak{G}p$; conversely such pairs vanish on every balanced tangent by the exact weak balance. $\square$

1.3. Finite-volume covariance first. For a regular finite source Θ define the exactly centered field

$$Z_\varepsilon = \sqrt{\mu_\varepsilon} (X_\varepsilon - DQ_\varepsilon(\Theta)), \quad X_\varepsilon = (\pi^\varepsilon, \Gamma^\varepsilon).$$

Theorem 1.4 (Projective Gaussian covariance). *For every finite family of weighted density/contact tests, the covariance matrices*

$$D^2 Q_\varepsilon(\Theta) = \mu_\varepsilon \text{Cov}_{\Theta, \varepsilon}(X_\varepsilon)$$

converge to a compatible positive semidefinite family Σ . All scaled cumulants of order at least three vanish, so the finite projections of Z_ε converge to $N(0, \Sigma)$. The null space is precisely the prepared conservation space plus the closed balance gauge.

Proof. The B2 normal complex-source expansion gives local uniform convergence of all finite derivatives. A connected k th cumulant of the scaled field is $O(\mu_\varepsilon^{1-k/2})$. Hence only the Hessian survives. Positivity and projective compatibility are inherited from finite covariance matrices. If a class has zero limiting variance, its pairing vanishes against every regular local tilted perturbation. Density of those perturbations in the balanced tangent core and Theorem 1.3 identify the class with the gauge plus the prepared constraints. $\square$

Theorem 1.5 (Second epi-derivative and covariance inverse). *Let I be the exposed joint rate and $x = DQ(\Theta)$. The second epi-derivative*

$$d_e^2 I(x \mid \Theta)(z) = \Gamma\text{-}\lim_{t \downarrow 0} \frac{2}{t^2} [I(x + tz) - I(x) - t\langle \Theta, z \rangle]$$

is the closed quadratic form

$$d_e^2 I(x \mid \Theta)(z) = \begin{cases} \langle z, \Sigma^\dagger z \rangle, & z \in \overline{\text{Ran } \Sigma}, \\ +\infty, & \text{otherwise.} \end{cases}$$

On the regular short-time balanced core it equals the closure of $\mathfrak{h}_{f,q}$ from Corollary 1.2.

Proof. On each finite projection, analytic convex duality gives the classical second-order conjugacy between the pressure Hessian and the second epi-derivative of its Legendre transform. Tilts $\Theta + th$ provide recovery sequences on the covariance range; separation makes the value infinite off its closure. Increasing the projections gives a monotone family of closed quadratic forms and hence a Mosco limit. For smooth balanced tangents, differentiating the actual action and using Theorem 1.1 yields exactly $\mathfrak{h}_{f,q}$; no additional balance curvature occurs. Coercivity makes this core dense in the form domain, proving equality of the closures. $\square$

1.4. Process tightness.

Lemma 1.6 (Localized connected-cumulant estimate). *For deterministic intervals $I_1, \dots, I_k$ and compactly supported weighted tests, the connected cumulant has an L^1 density in one anchor time and exponentially decaying relative-time tree distance. In particular,*

$$\mathbb{E}|Z_\varepsilon([s, t], F)|^4 \leq C|t - s|^2 + C\mu_\varepsilon^{-1}|t - s|, \quad \mathbb{E}|Z_\varepsilon([s, t], F)|^2 \leq C|t - s|.$$

Proof. Localize the B2 connected trajectory expansion by inserting the deterministic time indicators before summing graphs. Root each connected graph at its earliest contact. Every additional marked time is joined to the root by a real trajectory edge whose renewal density decays exponentially on the short-time source chart. Integrating the tree majorant gives an L^1 anchor density and the displayed second/fourth moment bounds, including intervals sharing an endpoint. Boundary atoms are already included in the flux trace measure and contribute the term of order $\mu_\varepsilon^{-1}|t - s|$. $\square$

Theorem 1.7 (Prepared density–contact Gaussian process). *The exactly centered microscopic fields converge in the weighted nuclear dual path space to the unique Gaussian balanced process with covariance Σ . Its Cameron–Martin space is the quotient form domain of Theorem 1.5; the initial covariance is the B1 microcanonical Schur complement.*

Proof. Finite-dimensional convergence is Theorem 1.4. Lemma 1.6 gives Aldous estimates on every member of a countable nuclear test core; Mitoma’s criterion gives tightness. The weak balance passes to the limit by closedness of B2’s Green graph. The limiting finite-dimensional laws and balance determine a unique Gaussian process. Theorem 1.5 identifies its Cameron–Martin form, and B1 identifies the prepared initial block. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: `qiqian@pku.edu.cn`